

SOFTWARE ANALYSIS AND DESIGN (SAD)

Professional Portfolio Project Report

Student Name: Chamika Gayantha (2022/ICTS/165)

Repository: <https://github.com/Chamika-Gayantha/chamika-portfolio>

Live Deployment: <https://chamikagayantha.vercel.app/>

Institution: Faculty of Technological Studies, University of Vavuniya

1. Problem, Objectives, Stakeholders, and Scope

1.1 Problem Statement

Traditional paper-based resumes do not effectively demonstrate an ICT student's practical software development skills, UI/UX abilities, and completed projects. Academic lecturers, recruiters, and industry professionals need an easy way to view a student's skills, projects, source code, and live applications. Therefore, this project focuses on developing a centralized, responsive, and professional personal portfolio website that presents academic information, technical skills, projects, and professional contact details in one place.

1.2 System Objectives

- **Centralize Profile:** Present academic projects, technical skills, education, and personal information in one location.
- **Responsive Accessibility:** Ensure the website works effectively on mobile phones, tablets, and desktop computers.
- **Interactive Showcase:** Provide access to GitHub repositories, live project demonstrations, and communication channels.
- **Professional Branding:** Create a professional online presence as an ICT undergraduate and aspiring software developer.

1.3 Key Stakeholders

Primary User: Chamika Gayantha

Target Users:

- Academic lecturers and assessors
- IT recruiters
- Software developers and technical leads
- Fellow students and peer developers

1.4 Project Scope

In-Scope:

- Single-page responsive portfolio website
- Personal profile and biography
- Education information
- Technical skills showcase
- Project showcase
- GitHub repository links
- Live project links
- Downloadable CV
- Contact section and social media links

Out-of-Scope:

- Complex database management
- User authentication
- Online payment processing
- Advanced backend administration

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

ID	Requirement Name	Description
FR-01	Hero & About Section	Displays a professional introduction, personal biography, and CV download option.
FR-02	Skill Matrix Display	Displays technical skills such as Java, Python, HTML, CSS, JavaScript, and UI/UX design.
FR-03	Project Showcase	Presents completed projects with descriptions, technologies used, GitHub links, and live demonstrations.
FR-04	Contact Channel	Provides contact options through a contact form and social media links.
FR-05	Navigation	Allows users to smoothly navigate between different sections of the portfolio.

2.2 Non-Functional Requirements

ID	Category	Specification
NFR-01	Usability	The website should provide simple and intuitive navigation.
NFR-02	Responsiveness	The layout should adapt properly to mobile, tablet, and desktop screen sizes.
NFR-03	Performance	The website should load quickly and provide a smooth user experience.
NFR-04	Availability	The website should remain accessible through its Vercel deployment.

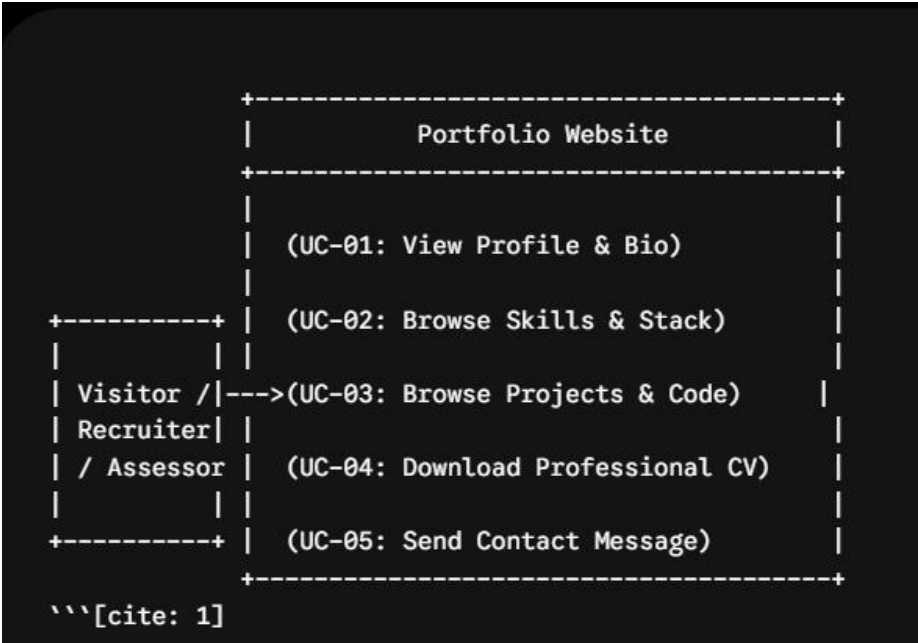
3. Use-Case Diagram and Descriptions

3.1 Use-Case Overview

The main actor of the system is the Visitor, which can include an academic assessor, recruiter, or other interested person.

The visitor can perform the following activities:

- UC-01 – View Profile & Bio
- UC-02 – Browse Skills & Technology Stack
- UC-03 – Browse Projects & Source Code
- UC-04 – Download Professional CV
- UC-05 – Send Contact Message



3.2 Key Use-Case Description

Use Case UC-03: Browse Projects & Source Code

Primary Actor: Academic Assessor / Recruiter / Visitor

Pre-condition: The portfolio website is successfully loaded in a web browser.

Main Success Scenario:

1. The visitor navigates to the Projects section.
2. The system displays the available projects.
3. Each project provides information such as the project name, description, and technologies used.
4. The visitor selects the GitHub repository or Live Demo option.
5. The system opens the selected repository or live application in a new browser tab.

Post-condition: The visitor successfully views the selected project's source code or live application.

4. Activity Diagram, Site Map, and Wireframes

4.1 Site Map Structure

Portfolio Home

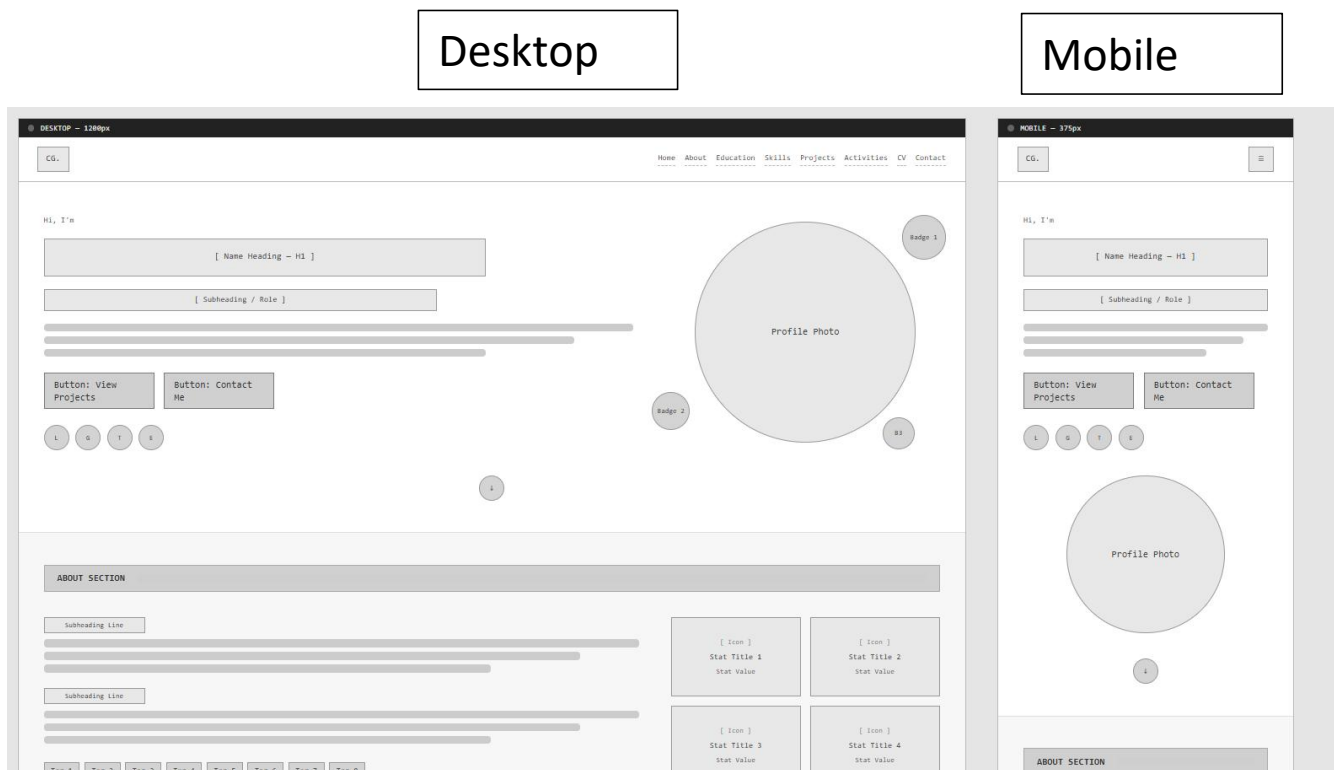


4.2 User Flow Activity

6. User Landing → The user opens the portfolio website and views the Hero section.
7. Navigation → The user selects a required section such as About, Skills, Projects, or Contact.
8. Project Review → The user opens the Projects section and selects a GitHub repository or live project.
9. External Redirect → The selected GitHub repository or live application opens in a new browser tab.
10. Contact Interaction → The user can complete the contact form or use the provided social media links to communicate.

4.3 Wireframes

The portfolio was designed with a simple and user-friendly layout. The main sections include the Hero section, About section, Skills section, Projects section, and Contact section.



5. Testing Evidence and Improvements

5.1 Test Execution Matrix

Test ID	Category	Test Action	Expected Outcome	Status
TC-01	Navigation	Click navigation menu links	The website smoothly navigates to the selected section.	PASS
TC-02	Responsiveness	Open the website on a mobile viewport	The layout adjusts correctly for smaller screens.	PASS
TC-03	External Links	Click a GitHub project link	The selected GitHub repository opens correctly.	PASS
TC-04	Form Validation	Submit the contact form with empty fields	The system identifies the missing required information.	PASS

5.2 Testing Improvements

Testing helped identify areas where the website could be improved. Navigation, responsiveness, external links, and form validation were checked to ensure that the website works correctly for different users and devices. The responsive design was particularly important because users may access the portfolio using different screen sizes.

6. Deployment and Working Links

The portfolio website was uploaded to GitHub for source-code management and deployed using Vercel for public access.

GitHub Repository: <https://github.com/Chamika-Gayantha/chamika-portfolio>

Live Portfolio Website: <https://chamikagayantha.vercel.app/>

Deployment Process

11. The portfolio source code was created and tested locally.
12. The project files were uploaded to GitHub.
13. The GitHub repository was connected to Vercel.
14. Vercel built and deployed the website.
15. The deployed website was tested using the public URL.

LinkedIn ScreenShot



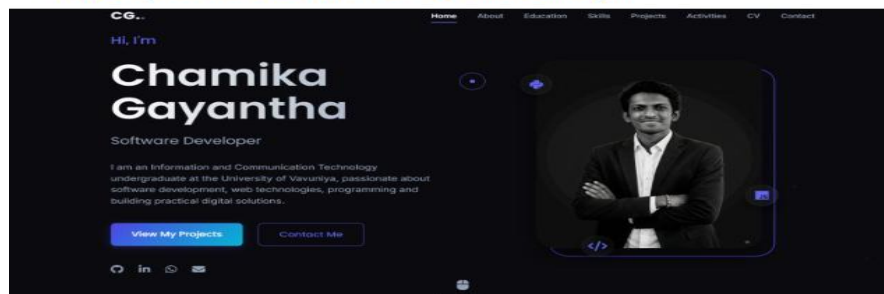
Chamika Gayantha • You
BICT(Hons) Undergraduate
9m • Edited •

🚀 My Personal Portfolio Website

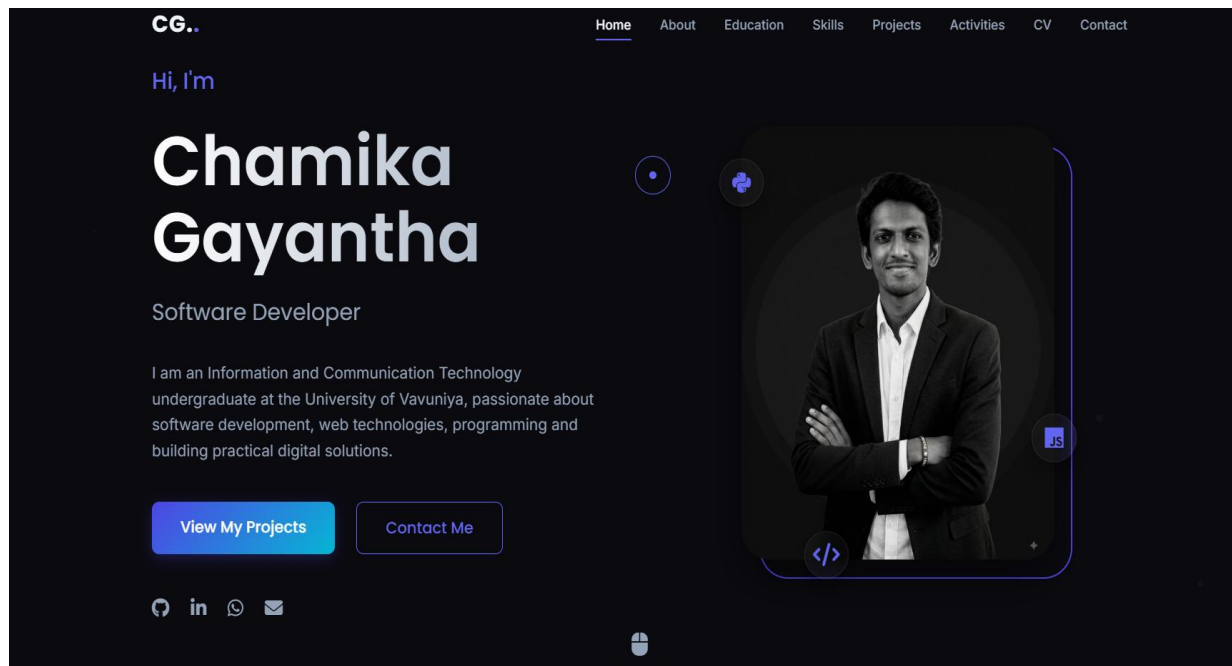
Excited to share my new personal portfolio website! 🚀💻
It showcases my projects, skills, education, and experience as an ICT undergraduate and software developer.

- 🛠️ Built with HTML, CSS & JavaScript
 - 🌐 Deployed using Vercel
 - 📄 Github Link: <https://lnkd.in/gwJ3mKSJ> ✓
 - 🌐 Live Portfolio: chamikagayantha.vercel.app ✓
- Always learning, building, and improving! 🚀

#Portfolio #WebDevelopment #SoftwareDevelopment #HTML #CSS
#JavaScript #GitHub #Vercel #ICT #StudentDeveloper



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7. Reflection on Career Value and SAD Learning

Developing this portfolio using System Analysis and Design principles helped transform a basic website into a more structured software project. By identifying the problem, defining system objectives, analysing stakeholders, specifying functional and non-functional requirements, and designing use-case and activity models, the development process became more organized.

The project also provided practical experience in web development, responsive design, GitHub-based version control, and cloud deployment using Vercel.

Most importantly, the portfolio provides a professional platform to demonstrate my technical skills and completed projects to lecturers, recruiters, and other professionals. This project helped me understand how System Analysis and Design concepts can be applied to a real-world software development project.

Conclusion

The Professional Portfolio Website successfully provides a centralized platform for presenting personal information, education, technical skills, projects, and professional contact details. The project demonstrates the practical application of System Analysis and Design concepts while also providing valuable experience in web development, version control, responsive design, and cloud deployment. The completed portfolio can be accessed through the live deployment: <https://chamikagayantha.vercel.app/>